Hybrid finite element and neural network methods for parametric PDEs

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MACS seminar
("Modélisation, Analyse et Calcul Scientifique")

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Context and motivation

TODO

Outline

TODO

Elliptic parametric PDEs

TODO

Classical numerical methods for PDEs

Discontinuous Galerkin method

TODO

Espace discontinu, flux numériques, pénalisation
Comparaison CG / DG

Continuous Galerkin method

TODO

Problème modèle, formulation faible
Espace $\mathbb{P}_k$ de Lagrange continu

Physics-informed neural networks

TODO

Principe, fonction de coût
Version paramétrique

1st hybridization

Enriched finite element method using PINNs

Presentation

Pipeline of the hybrid method

TODO

Additive approach

TODO

Convergence analysis

TODO

Numerical results

Test case 1
Test case 2

Numerical results

Test case 1

Test case 2

Test case 1

TODO

Numerical results

Test case 1

Test case 2

Test case 2

TODO

Conclusion, limitations and perspectives

Conclusion, limitations and perspectives

TODO

2nd hybridization

Learnable Galerkin methods

Introduction

Autodiff

TODO

Formalization

Optimization problem

TODO

Non-parametric enrichment (PINN loss)

Algorithm

TODO

Parametric enrichment (neural operator)

Algorithm

TODO

Conclusion

Conclusion

Contribution 1 : TODO

TODO

Contribution 2 : TODO

TODO

Perspectives :

- TODO

Appendix – 1st hybridization

Enriched finite element method using PINNs

Additional details

Additional details

TODO

Détails de la méthode (hors présentation principale)

Additional numerical results

Additional numerical results

TODO

Résultats numériques supplémentaires

Appendix – 2nd hybridization

Learnable Galerkin methods

Additional details

Additional details

TODO

Détails de la méthode (hors présentation principale)

Additional numerical results

Additional numerical results

TODO

Résultats numériques supplémentaires